# Brunhes Epoch Geomagnetic Secular Variation on Reunion Island

N. D. WATKINS

*Graduate School of Oceanography, University of Rhode Island*
*Kingston, Rhode Island 02881*

A paleomagnetic survey has been made of Brunhes epoch lavas on Reunion Island by using an average of more than seven separately oriented samples per body. The angular standard deviation of virtual geomagnetic poles (corrected for within-lava variation) is 17.7°, upper and lower confidence limits being 22.8° and 14.4°, respectively. An earlier survey involving an average of less than two separate samples per body (so that no within-lava correction was possible) produced corresponding confidence limits of 23.2° and 17.5°. Thus the contrasting sampling methods have not yielded distinguishable results. Both results are consistent with a Brunhes epoch geomagnetic secular variation for the region higher than present values in the same latitudes, which could be caused by an inadequate sampling of the paleosecular variation spectrum or by the existence of a regional secular variation anomaly. An incidental result of the survey is the discovery of a large excursion of the geomagnetic field at $t = 0.51$–$0.61$ m.y.

In an earlier paper [*Watkins and Nougier*, 1973] a study of the Brunhes epoch ($t = 0$–$0.69$ m.y.) geomagnetic secular variation in the Indian Ocean region was described by using paleomagnetic surveys of lavas from the Comores Islands, the Crozet Islands, and Amsterdam Island. Sufficient separate samples were taken from each lava that correction for within-lava dispersion could be applied to computation of the total angular dispersion of virtual geomagnetic poles, $S_T$, to arrive at a best measure of the real geomagnetic field secular variation $S_F$.

In this paper the survey is extended by using the same experimental and analytical methods to the Brunhes epoch lavas of Reunion Island (Figure 1). In terms of paleosecular variation studies, the island has added significance, since an earlier survey made by *Chamalaun* [1968*a*, *b*] was rejected by *Cox* [1969*a*, p. 258] on the grounds that the sampling method employed was inadequate for determination of $S_F$. The second survey of the Reunion Island lavas, using six or more widely spaced separately oriented samples per lava as preferred by *Doell and Cox* [1963, 1972], can therefore provide a means to compare $S_F$ with the value of $S_T$ provided by Chamalaun's results, which are based on an average of less than two separate samples per lava.

Copyright © 1973 by the American Geophysical Union.

## GEOLOGY OF REUNION ISLAND

Reunion Island has been very well studied geologically. *McDougall* [1971] has published the most recent summary of the island's stratigraphy, to which he has added a large number of K:Ar age determinations. Two large volcanic centers, Piton des Neiges and Piton de la Fournaise, dominate the island (Figure 1). Piton des Neiges is divided into an oceanite series [*Upton and Wadsworth*, 1965] of olivine basalts reaching 1000 meters in thickness overlain by the differentiated series of alkali andesite and feldsparphyric basalt of a similar maximum thickness. *McDougall's* [1971] results show that the oceanite series extends from about 0.4 to 2.1 m.y. in age, whereas both the differentiated series and the olivine basalts constituting Piton de la Fournaise extend from about 0.4 m.y. ago to the Recent.

Because of the K:Ar analyses, paleomagnetic results on the same basalts have been used to assist in definition of the geomagnetic polarity time scale [*Chamalaun and McDougall*, 1966], and, as was mentioned above, *Chamalaun* [1968*a*, *b*] was able to attempt paleomagnetic secular variation analyses for restricted age ranges.

## METHODS

*Field.* One hundred and sixty cores 2.5 cm in diameter were collected from a total of 22 separate lavas. The bodies were carefully

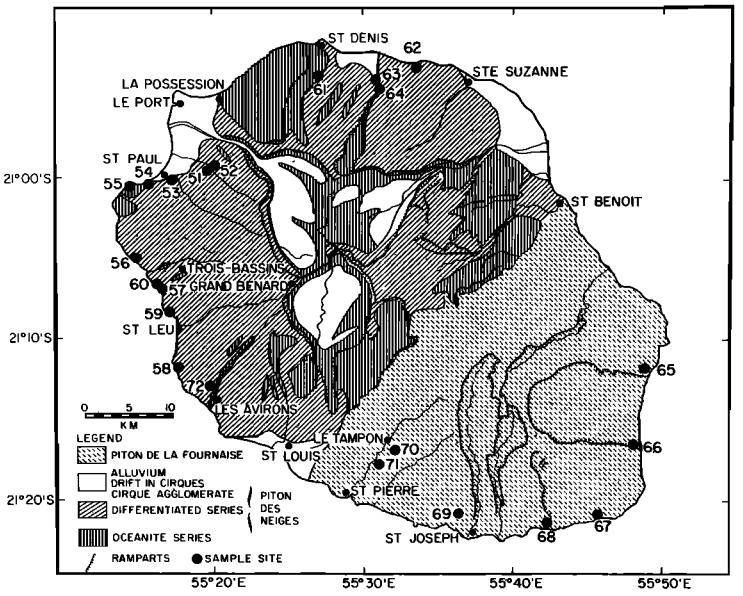



Fig. 1. Geological sketch map of Reunion Island, showing sample site locations [after *McDougall*, 1971, Figure 2].

selected to satisfy two criteria: restriction to the Brunhes epoch and locations facilitating precise geographic orientation of all cores. The restriction to the Brunhes epoch was accomplished by use of McDougall's K:Ar results together with known stratigraphy and field polarity measurements. The precise geographic orientation was accomplished by use of the excellent 1:100,000 topographic maps of the island, enabling each core to be oriented with respect to known geographic bearings. Within each lava, cores were separated laterally by an average of 5 meters. The location of each lava sampled is shown in Figure 1.

*Laboratory.* Specimens 2.2 cm in length were taken from each core. The remanent magnetism was measured by using spinner magnetometers. Unstable components were minimized by application of alternating magnetic fields of 100 and 200 oe. The mean directions of remanent magnetism for each lava were computed by using a minimum dispersion criterion. As reviewed by *Watkins et al.* [1972], the effect of this method is to weight directions to those in the most stable samples, which is clearly desirable. All data for each lava were analyzed by using methods described in detail by *Watkins and Nougier* [1973] for comparison with the $S_f$ versus latitude expressions of the conventional geomagnetic field models (Figure 2).

## RESULTS

One lava (Figure 1, lava 60) was found to possess remanent magnetism grossly modified by a lightning strike, so that no meaningful data could be obtained. Magnetic stability inferred from the rate of decrease of intensity of magnetization with higher peak demagnetizing field applications was found to be variable within all lavas examined, as may be expected from earlier detailed studies of single lavas [*Watkins and Haggerty*, 1967; *Wilson et al.*, 1968]. All other results for each lava are given in Table 1.

Two units (lavas 58 and 61) exhibit low-latitude southern hemisphere virtual geomagnetic poles (VGP). These lavas were inadvertently sampled as normal polarity units, since the natural remanent magnetism in the hand samples used for preliminary field examination apparently possessed some unstable normal polarity overprint.

In Tables 2–4 the methods and results of the relevant statistical analyses of data for all normal polarity lavas combined are defined and presented. *Chamalaun's* [1968*b*] results for the

21562202, 1973, 32, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB078i032p07763 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

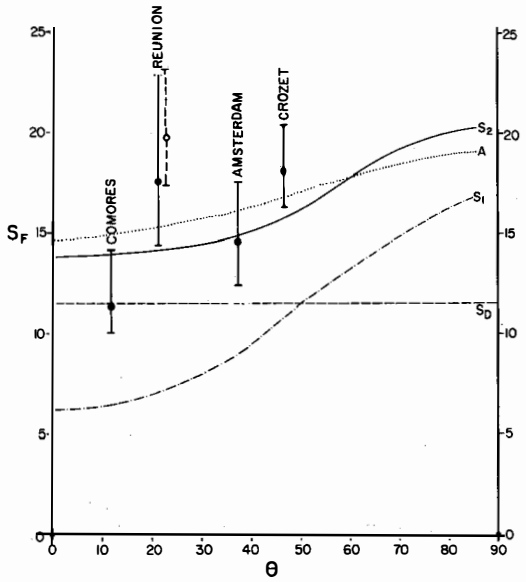



Fig. 2. Paleosecular variation of the geomagnetic field during the Brunhes epoch on Reunion Island and other Indian Ocean islands superimposed on the angular standard deviation of virtual geomagnetic poles, $S_F$, versus latitude values for various geomagnetic field values; $S_D$ is the effect of dipole wobble [*Creer et al.*, 1959]: the value of 11° is estimated by *Doell* [1969] to be typical of the Brunhes epoch; $S_1$ is the effect of the 1945 international geomagnetic reference field (IGRF) nondipole field averaged at 10° longitudinal intervals and between hemispheres; $A$ is the effect of a model comprising fixed amplitude but varying orientation vector (nondipole field) added to constant axial dipole field [*Irving and Ward*, 1964]; $S_2$ is the effect of total field behavior of the 1964 IGRF, equal to $(S_D^2 + S_1^2)^{1/2}$. Data for the Comores Islands, Amsterdam Island, and Crozet Island are from *Watkins and Nougier* [1973]. Ninety-five-percent confidence intervals [*Cox*, 1969*b*] are added. The Reunion Island result is from the present paper (Tables 2–4). The dashed line is the value of $S_T$ from the earlier survey of Reunion Island lavas by *Chamalaun* [1968*b*], as computed by *Watkins et al.* [1972]: the result is plotted at a slightly higher than true latitude for ease of comparison. For definition and method of computation of $S_F$ and $S_T$, see Table 4.

Brunhes lavas of Reunion Island are included for comparison.

## Discussion

*Low-latitude southern hemisphere VGP.* Lavas 58 and 61 both possess VGP in the lower latitudes of the southern hemisphere. It appears virtually certain that the remanent magnetism of lava 61 has recorded a geomagnetic departure during the early Brunhes epoch. Although McDougall has dated the body at $t = 0.58$ m.y. (Table 1), the corresponding paleomagnetic data were not presented by *Chamalaun* [1968*b*], conceivably because of their paleomagnetically anomalous character. This departure cannot be attributed uniquely to dipole or nondipole activity. *Watkins and Nougier* [1973] have discussed the published data on Brunhes epoch departures. None of the departures so far observed produce a VGP in the vicinity of that recorded by lava 61 (Tables 2–4). This is some evidence against a simple dipole tilting as the source of the data.

It is not impossible that lava 58 may be of Matuyama age ($t = 0.69$–$2.43$ m.y.), although the VGP latitude (Table 1) is not sufficiently high to be unequivocally of reversed polarity. It could therefore also represent a Brunhes epoch geomagnetic departure. The body is not dated isotopically. The ambiguity of the origin of the VGP for unit 58 is not uniquely resolvable.

*Geomagnetic secular variation.* Values of $S_F$ are presented in Table 4 and Figure 2. Results for similar determinations of $S_F$ from other latitudes in the southern Indian Ocean for the Brunhes epoch [*Watkins and Nougier*, 1973] are added to Figure 2. In addition, the result of analysis of *Chamalaun's* [1968*b*] data to produce $S_T$, as presented by *Watkins et al.* [1972], is shown for comparison adjacent to the Reunion Island result. The new survey of the Reunion Island lavas has not resulted in any significant reduction of the observed Brunhes epoch paleomagnetic secular variation for the island, except as the result of an increase in the 95% confidence interval (because of the fewer separate lavas involved).

The result has two possible implications. The similarity of $S_F$ and $S_T$ could be the result of an inadequate sampling of the secular variation spectrum in the present survey, so that the true average value of $S_F$ is not obtained. An inadequate sampling is more readily accomplished in practice in the case of a series of lavas extruded in rapid succession: here an unrealistically low value of $S_F$ would be obtained. As is shown in Table 1, however, the available K:Ar ages have a substantial range (of the

21562202, 1973, 32, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB078i032p07763 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Paleomagnetic Data for Each Lava

| Site No. (Figure 1) | $N$ | $D$ | $I$ | $R$ | $K$ | $\alpha_{95}$ | $dp$ | $dm$ | $\theta'$ | $\phi'$ | Age, m.y. | GA No. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 51 | 6 | 13.7 | -13.5 | 5.793 | 24.2 | 13.90 | 7.2 | 14.2 | 70.5 | 100.2 | | |
| 52 | 5 | 15.8 | -47.6 | 4.986 | 292.3 | 4.48 | 3.8 | 5.8 | 73.9 | 176.2 | 0.233 ± 0.005 | 1257 |
| | | | | | | | | | | | 0.225 ± 0.004 | |
| 53 | 6 | 58.2 | -23.7 | 5.798 | 24.7 | 13.74 | 7.8 | 14.7 | 33.9 | 146.3 | | |
| 54 | 8 | 0.0 | -29.0 | 7.972 | 254.2 | 3.48 | 2.1 | 3.8 | 84.3 | 55.6 | 0.440 ± 0.030 | 1293 |
| | | | | | | | | | | | 0.467 ± 0.042 | |
| 55 | 9 | 354.6 | -48.5 | 8.959 | 194.3 | 3.70 | 3.2 | 4.9 | 80.4 | 256.2 | 0.435 ± 0.009 | 1295 |
| 56 | 8 | 11.7 | -55.3 | 7.971 | 238.5 | 3.59 | 3.6 | 5.1 | 72.2 | 203.0 | | |
| 57 | 8 | 356.8 | -57.4 | 7.973 | 260.2 | 3.44 | 3.7 | 5.0 | 73.0 | 244.1 | | |
| 58 | 6 | 223.6 | -14.7 | 5.487 | 9.8 | 22.58 | 11.9 | 23.1 | -38.5 | 296.3 | | |
| 59 | 7 | 348.9 | -34.1 | 6.990 | 582.0 | 2.50 | 1.6 | 2.9 | 79.3 | 337.0 | | |
| 61 | 6 | 221.5 | -68.8 | 5.890 | 45.5 | 10.03 | 14.4 | 17.0 | -8.2 | 259.6 | 0.554 ± 0.025 | 1261 |
| | | | | | | | | | | | 0.602 ± 0.008 | |
| 62 | 7 | 7.5 | -46.7 | 6.921 | 76.4 | 6.95 | 5.8 | 9.0 | 80.4 | 191.8 | | |
| 63 | 7 | 329.3 | -36.1 | 6.990 | 611.9 | 2.44 | 1.7 | 2.8 | 61.3 | 322.1 | 0.291 ± 0.007 | 1294 |
| 64 | 6 | 337.3 | -50.7 | 5.953 | 106.2 | 6.53 | 5.9 | 8.8 | 67.3 | 294.0 | | |
| 65 | 9 | 8.2 | -35.5 | 8.948 | 152.7 | 4.18 | 2.8 | 4.8 | 82.2 | 135.5 | | |
| 66 | 8 | 354.3 | -26.9 | 7.945 | 127.1 | 4.93 | 2.9 | 5.4 | 81.2 | 16.6 | | |
| 67 | 9 | 353.8 | -31.1 | 8.988 | 685.8 | 1.97 | 1.2 | 2.2 | 82.6 | 1.3 | | |
| 68 | 8 | 332.9 | -34.9 | 7.961 | 181.1 | 4.13 | 2.7 | 4.8 | 64.5 | 324.9 | | |
| 69 | 8 | 7.8 | -40.3 | 7.967 | 210.9 | 3.82 | 2.8 | 4.6 | 82.5 | 160.4 | | |
| 70 | 8 | 13.5 | -31.8 | 7.905 | 74.0 | 6.48 | 4.1 | 7.3 | 76.6 | 130.4 | 0.327 ± 0.010 | 1290 |
| 71 | 6 | 316.8 | -56.7 | 5.863 | 36.4 | 11.25 | 11.8 | 16.3 | 49.5 | 292.3 | | |
| 72 | 8 | 10.9 | -27.6 | 7.944 | 124.7 | 4.98 | 3.0 | 5.4 | 77.8 | 114.9 | | |

The meanings of the symbols are as follows: $N$, total number of separately oriented samples for a given site; $D$ and $I$, mean declination and inclination of remanent magnetism in degrees east of north and below horizontal, respectively (obtained from unit vector per sample); $R$, resultant vector (unit vector per sample); $K = (N - 1)/(N - R)$; $\alpha_{95}$, semivertical angle of 95% confidence cone [*Fisher*, 1953]; $dp$ and $dm$, major and minor axes of 95% confidence oval about mean pole position in degrees along and normal to great circle from mean sample site location to mean pole position, respectively; $\theta'$ and $\phi'$, latitude and longitude of mean virtual geomagnetic pole, obtained by using unit vector per sample. All paleomagnetic data are for specimens treated in alternating demagnetizing fields of 100 or 200 oe. The age is the K:Ar age determination by *McDougall* [1971].

21562202, 1973, 32, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB078i032p07763 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Analysis of Directions (Reunion Island)

| | $B$ | $N$ | $D$ | $I$ | $R$ | $K$ | $\alpha_{95}$ | $\theta'$ | $\phi'$ | $dp$ | $dm$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (i) | 19 | 141 | 0.6 | -40.3 | 17.826 | 15.3 | 8.9 | 88.2 | 217.7 | 6.4 | 10.7 |
| (ii) | 18 | 135 | 357.2 | -40.2 | 17.209 | 21.5 | 7.6 | 86.9 | 290.9 | 5.6 | 9.2 |
| (iii) | 49 | 59 | 6.3 | -34.2 | 46.126 | 16.6 | 5.1 | 84.1 | 156.2 | 3.7 | 6.1 |

Symbol $B$ is the number of separate sites sampled; $K = (B - 1)/(B - R)$; the remaining symbols are as defined in Table 1. Row (i) is all data, row (ii) is data excluding units where $\theta' \leq 45°$ (Table 1), and row (iii) is the result from an earlier survey by *Chamalaun* [1968*b*].

order of 0.4 m.y.), which is more than half of the duration of the Brunhes epoch. The lavas sampled on the east coast (units 65–68) are almost certainly less than 0.1 m.y. in age, the age range of the lavas studied thus being increased. Therefore, although only 18 high-latitude VGP lavas have been studied, maximum prospects exist for an adequate sampling of the secular variation being presented for such a given number of separate bodies. If this is the case, it would follow that the earlier result by *Chamalaun* [1968*b*] (Figure 2) has also produced an acceptable result, and therefore the two sampling systems are equally effective. Thus the six or more separately oriented well-spaced cores are not necessary for paleosecular variation studies.

Alternatively, since both results have values of $S_F$ or $S_T$ significantly higher than those due to the present field (Figure 2), it can be argued that both surveys have inadequately sampled the secular variation, that a regional secular variation anomaly is present, or that (on the one hand) the high value of $S_T$ for the original value is due to a sampling of each lava that is inadequate to compute $S_F$, and (on the other hand) the high value of $S_F$ for the present survey is due to the sampling of an inadequate number of separate lavas, even though a substantial time range was involved. The preferred, if not the only, interpretation for the similarity of the results from the two surveys is the last possibility. Therefore the present ambiguities could probably be resolved by high-density sampling of the 49 lavas involved in the original survey.

The only definite conclusion is that the second survey has not produced a reduction in the Brunhes epoch secular variation value for Reunion Island.

TABLE 3. Analysis of Virtual Geomagnetic Poles (Reunion Island)

| | $B$ | $R$ | $K$ | $\alpha_{95}$ | $\theta'$ | $\phi'$ |
|---|---|---|---|---|---|---|
| (ii) | 18 | 17.18 | 20.7 | 7.3 | 86.0 | 282.3 |
| (iii) | 49 | 46.18 | 17.0 | 4.8 | 83.2 | 183.9 |

Symbols are as defined in Tables 1 and 2.

TABLE 4. Angular Standard Deviation of Virtual Geomagnetic Poles (Reunion Island)

| | $S_T$ | $K_T$ | $K_\omega$ | $K_A$ | $S_F$ | $Su$ | $Si$ |
|---|---|---|---|---|---|---|---|
| (iim) | 18.0 | 20.3 | 113.1 | 2100 | 17.7 | 22.8 | 14.4 |
| (iig) | 18.5 | 19.3 | 113.1 | 2100 | 18.2 | 23.5 | 14.8 |
| (iiim) | 19.9 | | | | | 23.2 | 17.5 |
| (iiig) | 21.1 | | | | | 24.5 | 18.6 |

The meaning of the symbols are as follows: $S_T$, total dispersion, equal to $(B - 1)^{-1/2}\Sigma_{i=1}^{B}(\delta i)^{1/2}$, where $\delta i$ is the angular difference between the $i$th VGP and the reference pole and $B$ is the number of sites; $K_T = 2(S_T)^{-2}$; $K\omega$, within-lava precision parameter for VGP, equal to $\Sigma_{i=1}^{N}[(N_i - 1)/(N_i - R_i)]$, where $N$ is the number of samples in the $i$th lava and $R$ is the resultant vector for the $i$th lava, the unit vector being applied to the VGP for each sample; $K_A$, precision parameter for effect of magnetic anomaly existing during cooling of lava (see *Doell* [1972] for discussion); $K_F$, precision parameter reflecting ancient geomagnetic field variation from $1/K_F = (1/K_T) - (1/\bar{N}K_\omega) - (1/K_A)$, where $\bar{N}$ is the average number of cores per lava; $S_F = (2/K_F)^{1/2}$; and $Su$ and $Si$, upper and lower confidence parameters for $S_F$ after *Cox* [1969*b*]. All data are for bodies with $\theta' \geq 45°$. The suffixes m and g refer to the reference pole for computation of $S_T$ (m, with respect to mean VGP; g, with respect to present geographic pole).

*Acknowledgments.* Dr. I. McDougall generously provided a finely detailed map showing the location of the lavas that were isotopically dated. Thanks are due to Dr. J. Nougier for assistance in the field. The field collection was made en route to an oceanographic cruise in the southern Indian Ocean, which was supported by grant GV32480 from the National Science Foundation. This support is gratefully acknowledged.

## REFERENCES

Chamalaun, F. H., Test of the Creer-Cox paleosecular variation model, *Earth Planet. Sci. Lett.*, *3*, 355–356, 1968*a*.

21562202, 1973, 32, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB078i032p07763 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chamalaun, F. H., Paleomagnetism of Reunion Island and its bearing on secular variation, *J. Geophys. Res., 73,* 4647–4659, 1968*b*.

Chamalaun, F. H., and I. McDougall, Dating geomagnetic polarity epochs in Reunion, *Nature, 210,* 1212–1214, 1966.

Cox, A., A paleomagnetic study of secular variation in New Zealand, *Earth Planet. Sci. Lett., 6,* 257–267, 1969*a*.

Cox, A., Confidence limits for the precision parameter *K, Geophys. J., 17,* 545–549, 1969*b*.

Creer, K. M., E. Irving, and A. E. M. Nairn, Palaeomagnetism of the Great Whin Sill, *Geophys. J., 2,* 306–323, 1959.

Doell, R. R., Paleomagnetism of the Kauai volcanic series, *J. Geophys. Res., 74,* 4857–4866, 1969.

Doell, R. R., Palaeomagnetic studies of Icelandic lava flows, *Geophys. J. Roy. Astron. Soc., 26,* 459–479, 1972.

Doell, R. R., and A. Cox, The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows, *J. Geophys. Res., 68,* 1997–2009, 1963.

Doell, R. R., and A. Cox, The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, in *The Nature of the Solid Earth,* edited by E. C. Robertson, pp. 245–284, McGraw-Hill, New York, 1972.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, Ser. A, 217,* 295–305, 1953.

Irving, E., and M. A. Ward, A statistical model of the geomagnetic field, *Pure Appl. Geophys., 57,* 25–30, 1964.

McDougall, I., The geochronology and evolution of the young volcanic island of Reunion, Indian Ocean, *Geochim. Cosmochim. Acta, 35,* 261–288, 1971.

Upton, B. G. J., and W. J. Wadsworth, The basalts of Reunion Island, Indian Ocean, *Bull. Volcanol., 29,* 7–24, 1965.

Watkins, N. D., and S. E. Haggerty, Primary oxidation variation and petrogenesis in a single lava, *Contrib. Mineral. Petrol., 15,* 251–271, 1967.

Watkins, N. D., and J. Nougier, Excursions and secular variation of the Brunhes epoch geomagnetic field in the Indian Ocean region, *J. Geophys. Res., 78,* 6060–6068, 1973.

Watkins, N. D., A. Hajash, and C. E. Abranson, Geomagnetic secular variation during the Brunhes epoch in the Indian and Atlantic ocean regions, *Geophys. J., 28,* 1–25, 1972.

Wilson, R. L., S. E. Haggerty, and N. D. Watkins, Variation of paleomagnetic stability and other parameters in a vertical traverse of a single Icelandic lava, *Geophys. J. Roy. Astron. Soc., 16,* 79–96, 1968.

(Received November 6, 1972;
revised July 12, 1973.)

21562202, 1973, 32, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB078i032p07763 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License